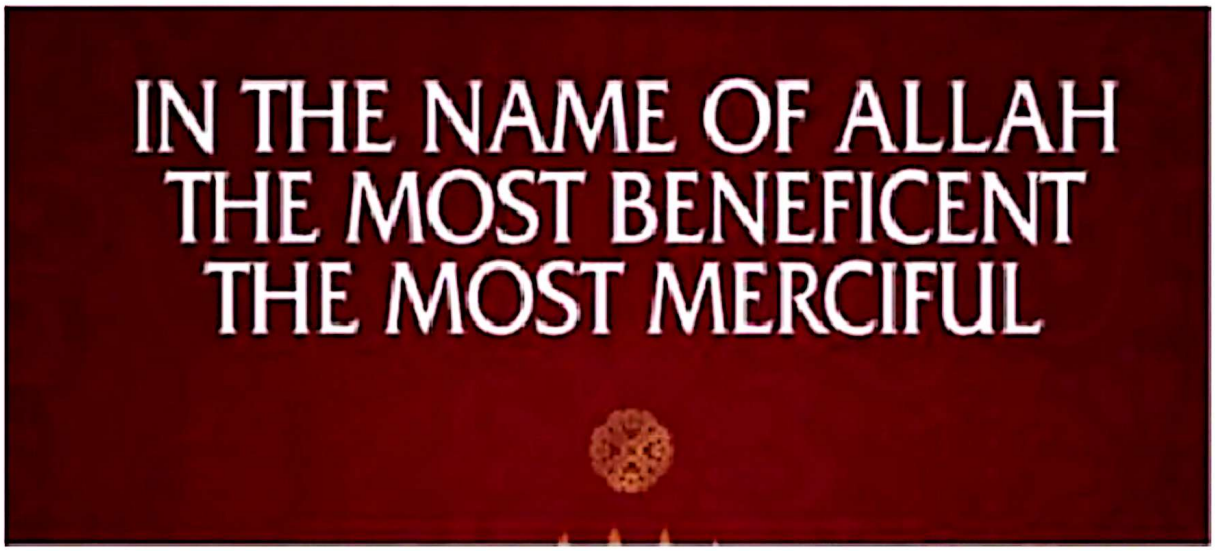
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الصلوٰۃ والسلام علیک یا رسول اللہ
وعلیٰ آلک واصحابک یا حبیب اللہ

Lecture No 1

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Mechanics

Mechanics is the branch of physical Sciences which studies the action of force on bodies. Under the action of these forces, bodies are in state of rest or of motion relative to some frame of reference.

Mechanics is further divided into two areas

Statics

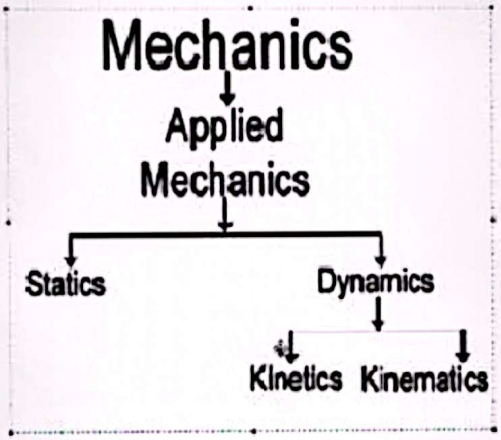
Statics is that branch of mechanics which deals with forces acting on a body which is at rest.

Dynamics

Dynamics is that branch of mechanics which accelerated motion.

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Some basic definitions

Body
Matter which supllies limited portion of space is called a body.
Mass
The quantity of a matter in a body is called its mass.
Particle
A body of negligibly small dimension is termed a particle.
Rigid body
A body is said to a rigid body if the distance between any two particles of the body remains unchanged so that the body does not expand or contract or change its shape in any way.
Rest and motion
A body is said to be in motion if, in the course of time, its position changes with respect to its surroundings, otherwise the body is said to be in rest. Rest and motion are relative concepts and are meaningful only when referred to surrounding bodies.

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Inertial and Non-Inertial Reference Frames

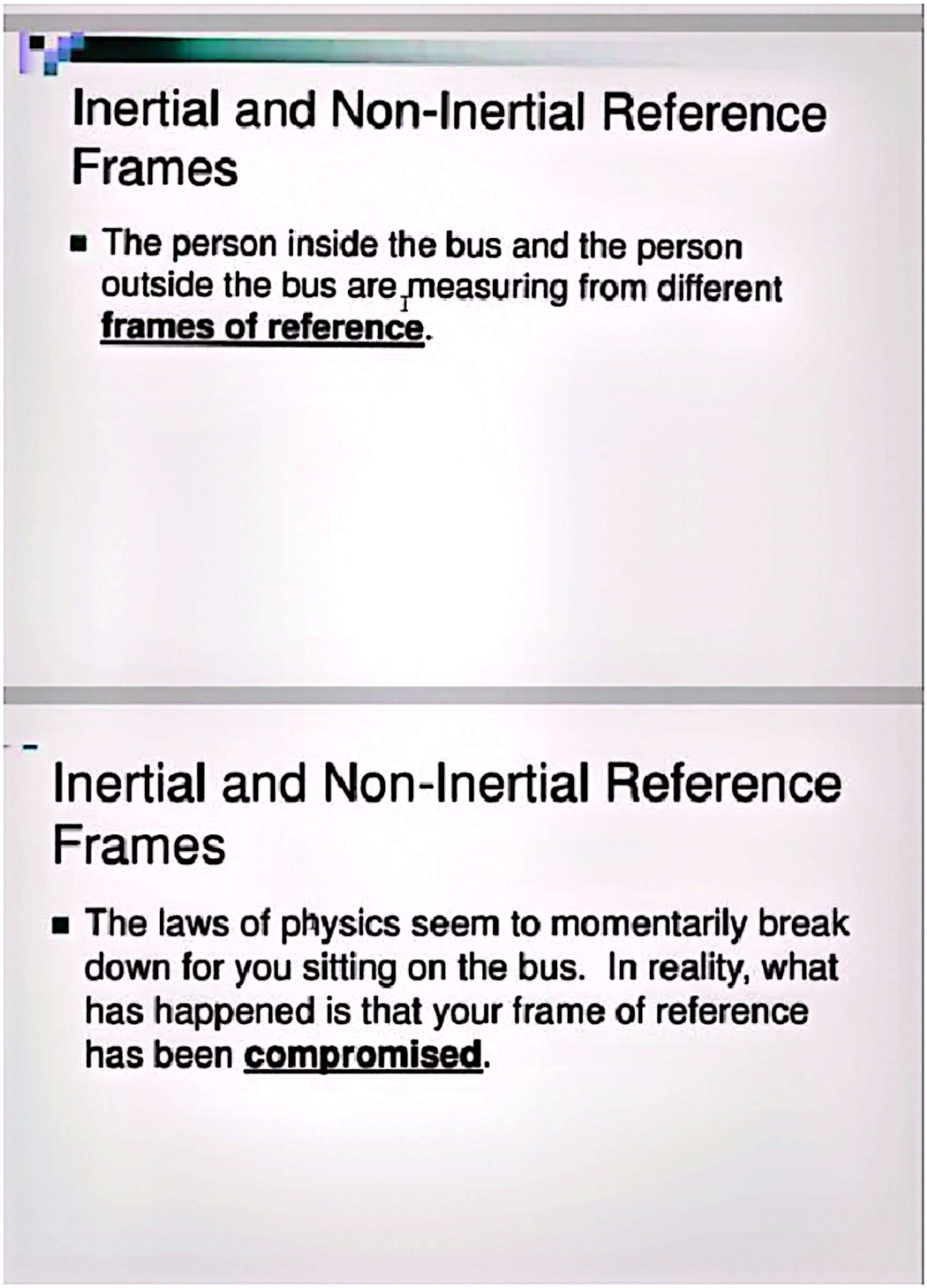
- Recall: Newton's first law of motion (law of inertia):

  An object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.

Inertial and Non-Inertial Reference Frames

- We've already talked about frames of reference in this course when we talked about relative velocity. You know that a person sitting on a bus watching a ball roll across the ground will have a very different measurement of the ball's velocity than a person standing outside the bus would measure the ball's velocity.

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Inertial and Non-Inertial Reference Frames

- An **inertial frame of reference** is a frame of reference in which the law of inertia and other physics laws **are valid**. Any frame moving at a constant velocity relative to another frame is also an inertial frame of reference.

Inertial and Non-Inertial Reference Frames

- When the breaks are applied to the bus, the bus undergoes a negative acceleration. At this moment, it becomes a non-inertial frame of reference.
- A **non-inertial frame** of reference is a reference frame in which **the law of inertia does not hold**.

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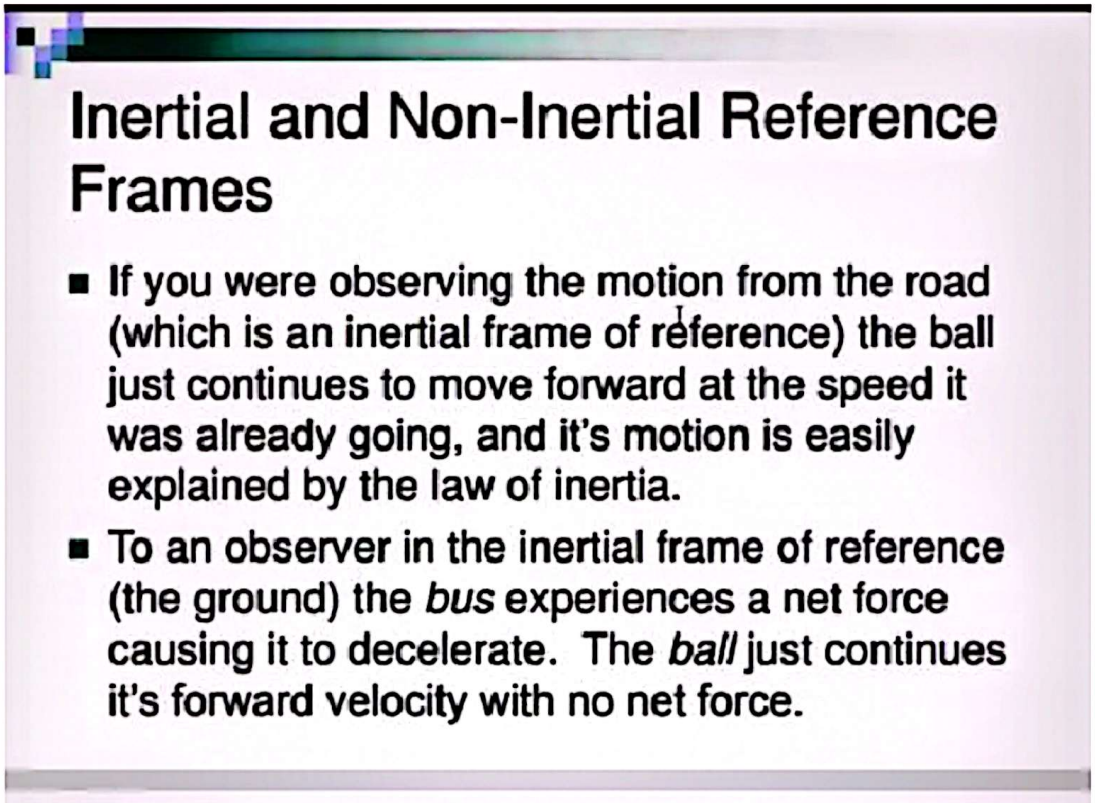
Inertial and Non-Inertial Reference Frames

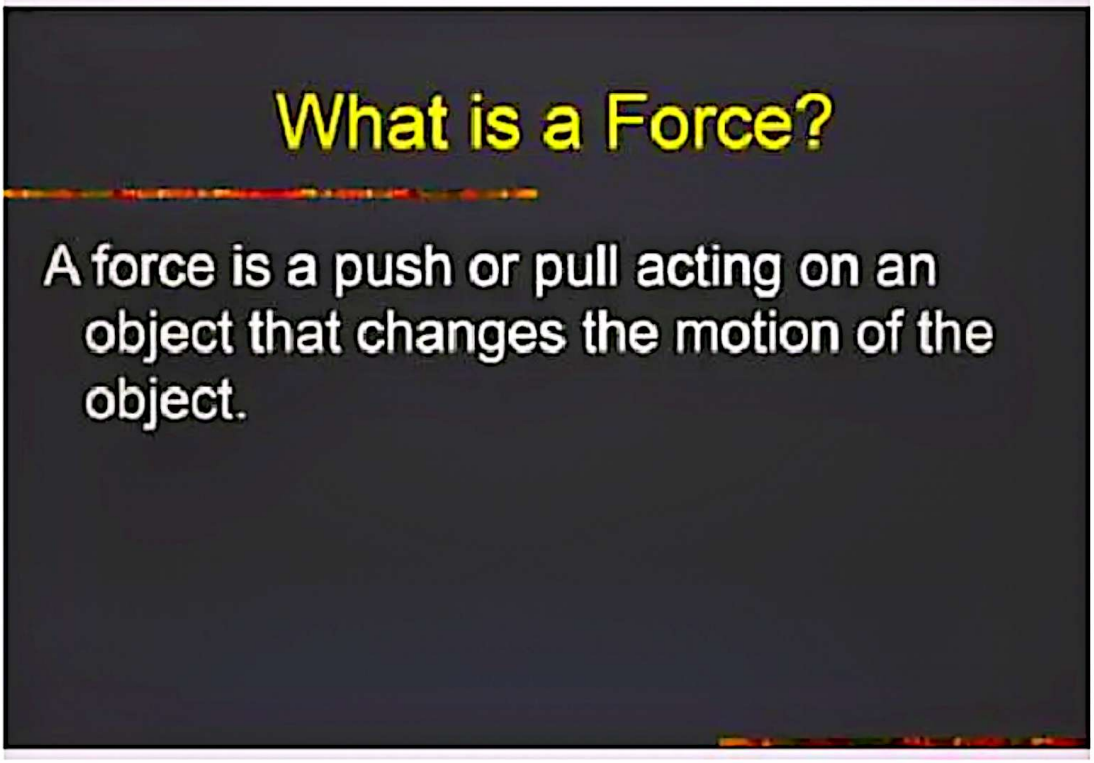
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Inertial and Non-Inertial Reference Frames

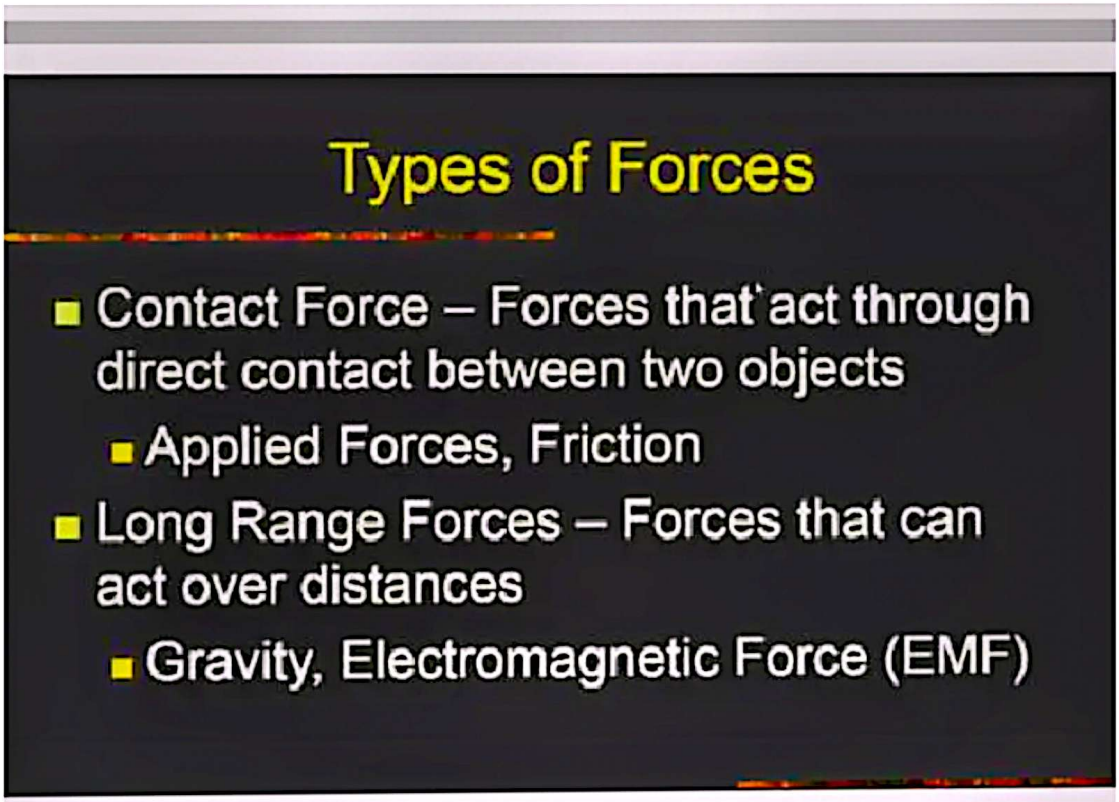
- Although the ball accelerates toward the front of the bus, there is **no net force** causing the acceleration.
- But if you are sitting on the bus, you observe the ball accelerating forward. That would imply to you as you sit on the bus that there is a **net force forward** on the ball.
- The reason there appears to be a net force on the ball is that you are observing the motion of the ball in the non-inertial reference frame.

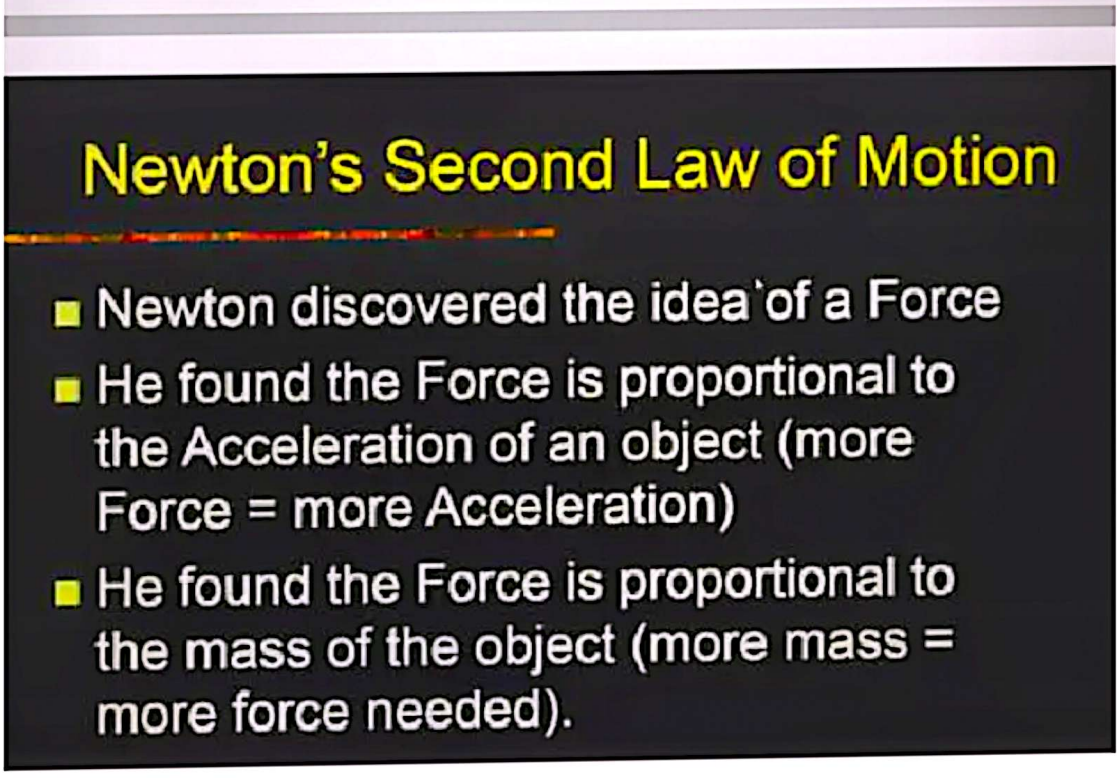
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Newton's Second Law of Motion

- Moving objects accelerate when an unbalanced force (F) acts on them. The stronger the force, the greater the acceleration (a). Also, the greater the mass (m) the greater the force required to change the motion.
- Force = mass x acceleration
  - F = ma

Newton's Second Law of Motion

- Force is a Vector Quantity and therefore has magnitude and direction.
- The direction of the force is the same as the direction of the acceleration.

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S.I. Unit For Force

- The Unit for Force is a Newton (N)
  - $1N = 1kg\ m/s^2$
  - A Newton (N) is defined as the amount of force required to accelerate 1 kg of mass at a rate of 1 m/s^2.

Net Force

- When Multiple forces are acting on an object. The Net Force is the amount of force that is left after adding all the forces on the object.

- Net force (F_{NET}) = Resultant Force (F_R)

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Balanced Forces

Balanced Forces are forces that are equal and opposite so that they cancel out.

10 N East → 10 N West ←

Net Force = +10 N + -10 N = 0

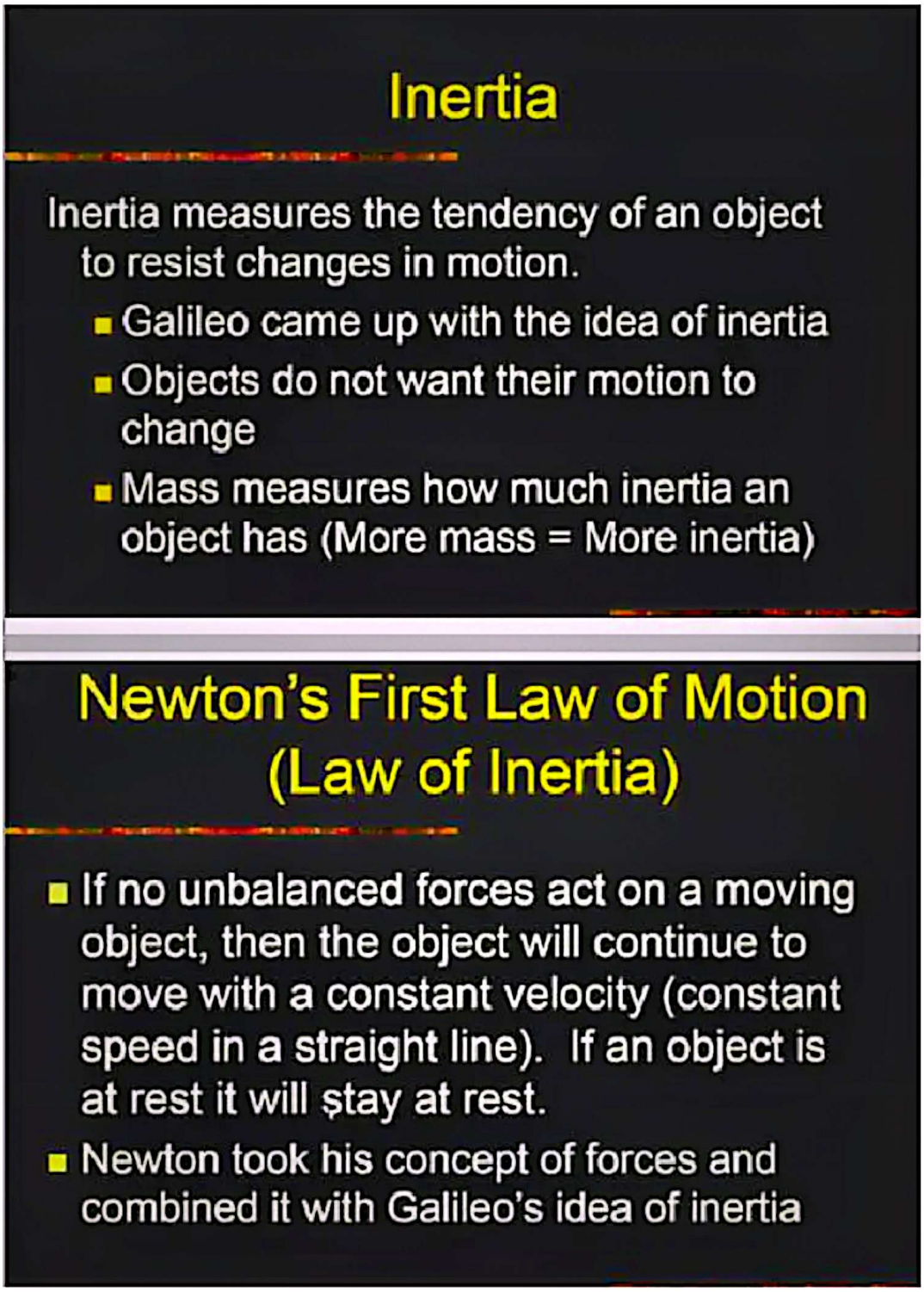
Unbalanced Forces

Unbalanced Forces are forces that when added do not cancel out and cause a change in the motion of the object.

30 N East → 10 N West ←

F_{net} = +30 N + -10 N = +20 N East

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Equilibrium

- First Condition for Equilibrium
  - If the Net Force acting on the object is zero

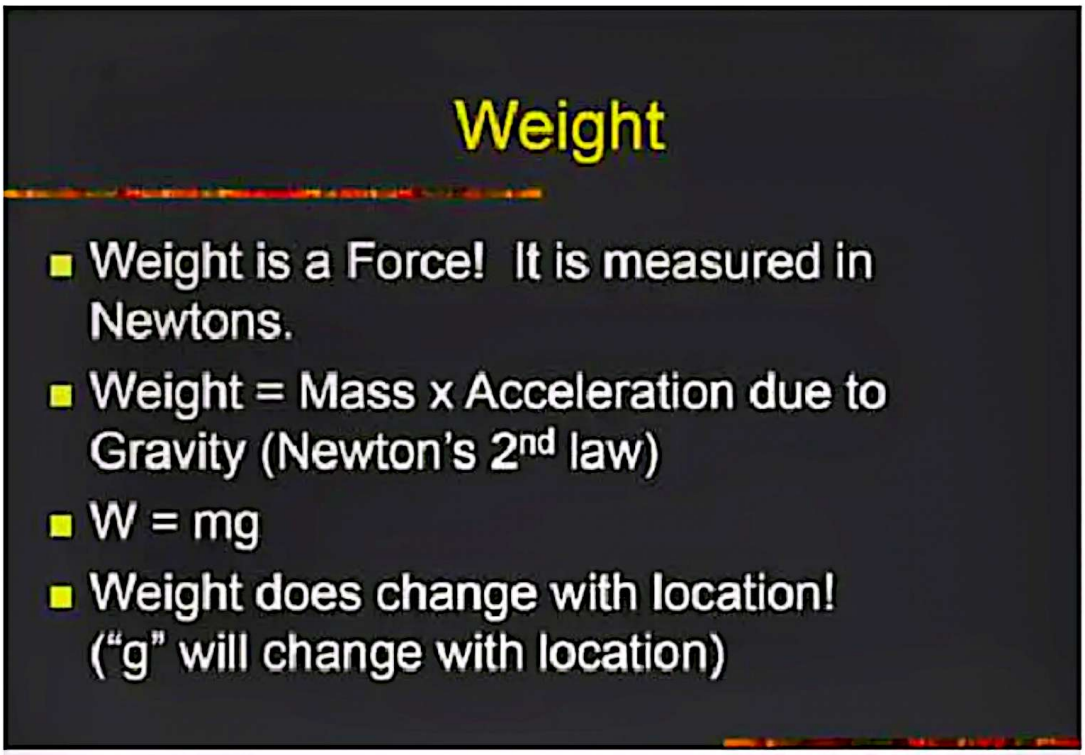
    $F_{NET} = 0 \quad a = 0$
  - The object is either stationary ($v = 0$) or traveling with a constant velocity (v = constant)

Mass and Weight

Mass is amount of matter that an object possess. Mass does not change with location.

Weight is the gravitational force that a large body (such as a planet) exerts on another object.

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Point of Application of a Force
The point of the body where the force acts is called the point of application.

Line of action of a Force
The line passing through the point of application in the direction of the force is called its line of action.

Force System
A set of forces acting on a particle or rigid body is called a force system or system of forces.

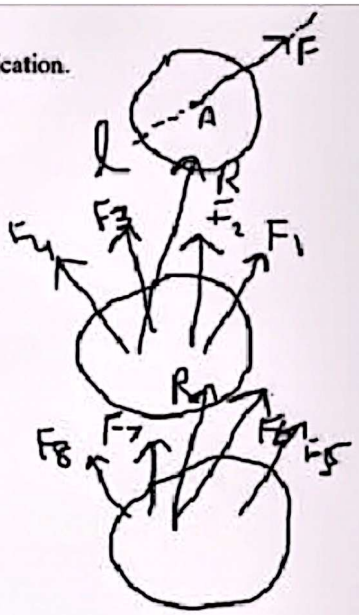
Equivalent force system
Two force system are said to be equivalent if one sysem can be replaced by the other without disturbing the instataneous state of rest or motion of a body.

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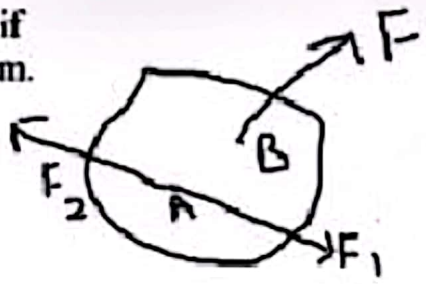
Law of Superposition
The action of a given system of forces on a body remains unchanged if we add to or subtract from them another system of forces in equilibrium.

Principle of Transmissibility of Forces
The point of application of a force, acting on a rigid body can be transferred to any other point of the body lying on the line of action of the force without changing the effect.

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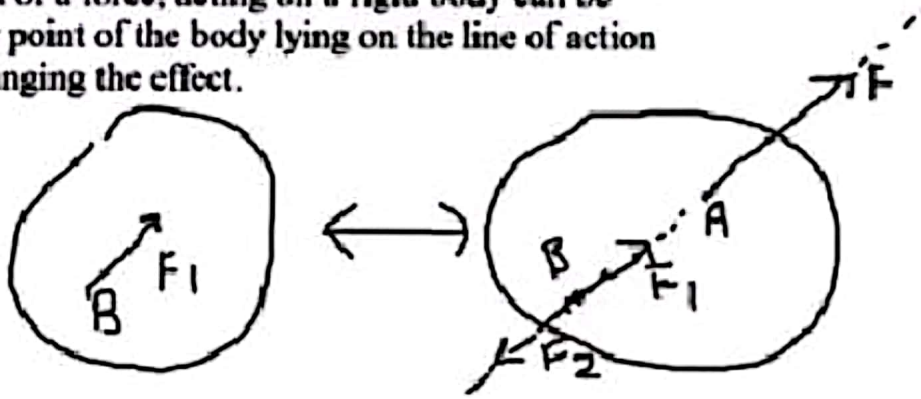
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VECTOR

The term vector is used by scientists to indicate a quantity (such as displacement or velocity or force) that has both magnitude and direction.

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REPRESENTING A VECTOR

A vector is often represented by an arrow or a directed line segment.

- The length of the arrow represents the magnitude of the vector.
- The arrow points in the direction of the vector.

DENOTING A VECTOR

We denote a vector by either:

- Printing a letter in boldface (**v**)
- Putting an arrow above the letter ($\vec{v}$)

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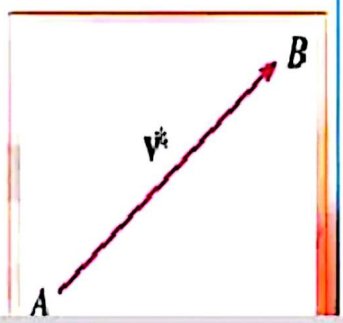
VECTORS

For instance, suppose a particle moves along a line segment from point *A* to point *B*.

VECTORS

The corresponding displacement vector **v** has initial point *A* (the tail) and terminal point *B* (the tip).

- We indicate this by writing $\mathbf{v} = \overrightarrow{AB}$.



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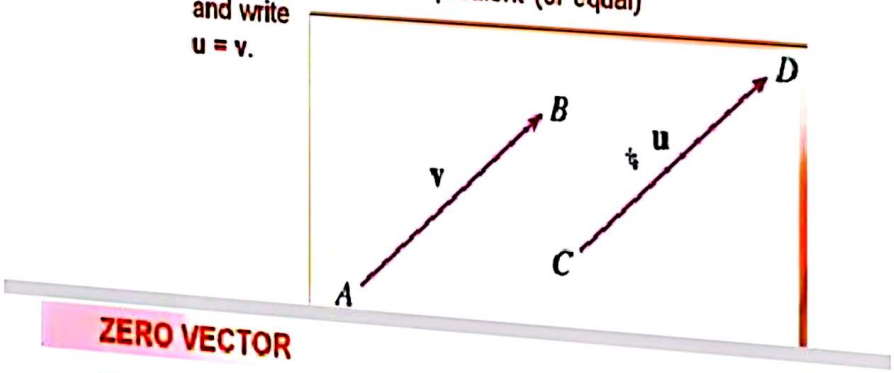
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VECTORS

Notice that the vector $\mathbf{u} = \overrightarrow{CD}$ has the same length and the same direction as **v** even though it is in a different position.

- We say **u** and **v** are equivalent (or equal) and write **u = v**.



ZERO VECTOR

The zero vector, denoted by **0**, has length 0.

- It is the only vector with no specific direction.

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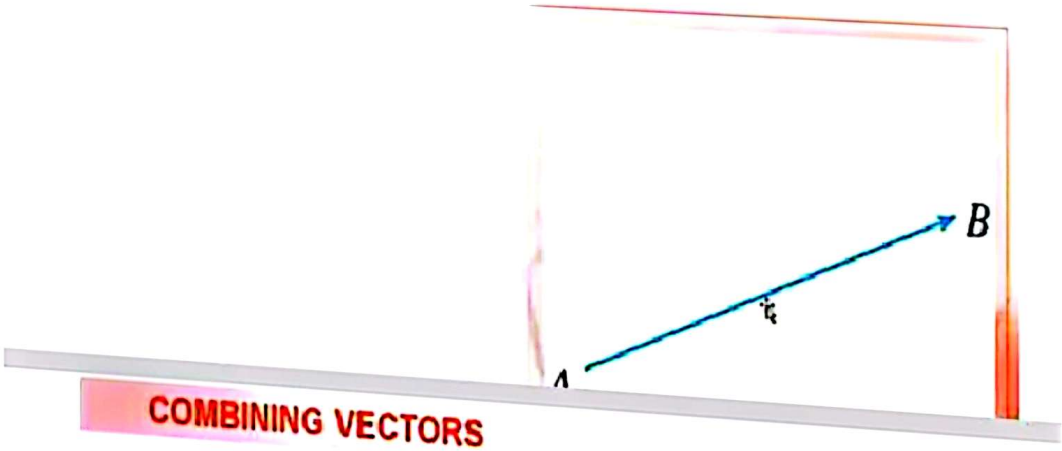
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COMBINING VECTORS

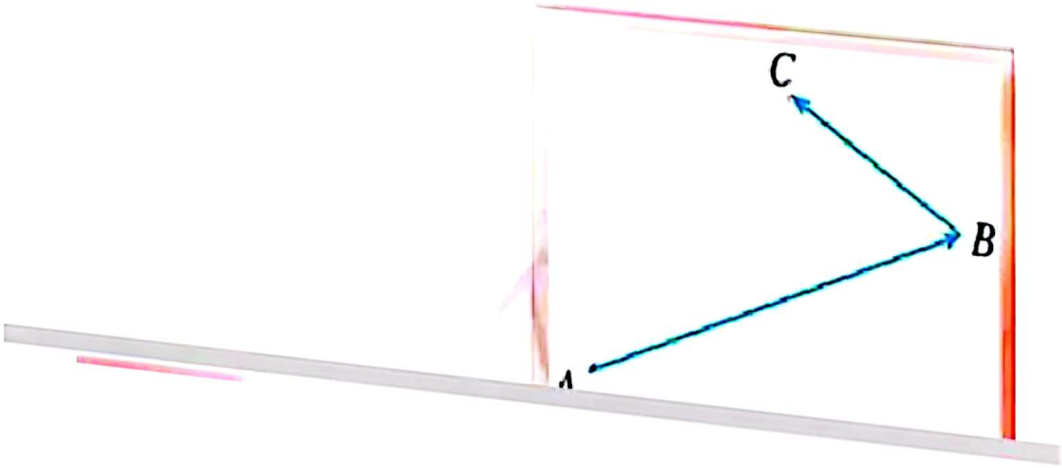
Suppose a particle moves from A to B.

So, its displacement vector is $\overrightarrow{AB}$.



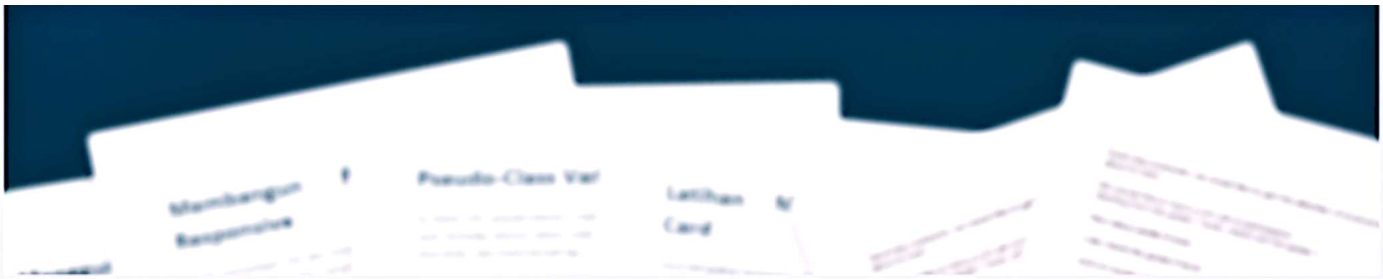
COMBINING VECTORS

Then, the particle changes direction, and moves from B to C—with displacement vector $\overrightarrow{BC}$.



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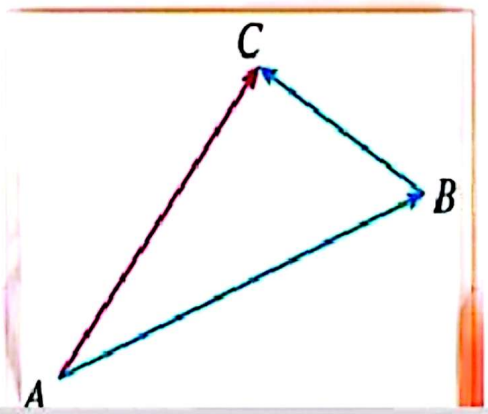
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COMBINING VECTORS

The resulting displacement vector $\overrightarrow{AC}$ is called the sum of $\overrightarrow{AB}$ and $\overrightarrow{BC}$.

We write:

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$



ADDING VECTORS

In general, if we start with vectors **u** and **v**, we first move **v** so that its tail coincides with the tip of **u** and define the sum of **u** and **v** as follows.

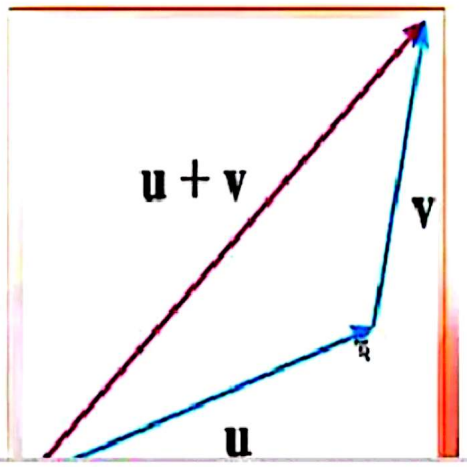
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VECTOR ADDITION—DEFINITION

If **u** and **v** are vectors positioned so the initial point of **v** is at the terminal point of **u**, then the sum **u** + **v** is the vector from the initial point of **u** to the terminal point of **v**.

VECTOR ADDITION

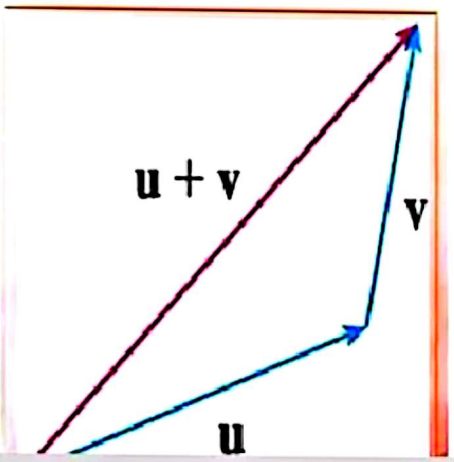
The definition of vector addition is illustrated here.



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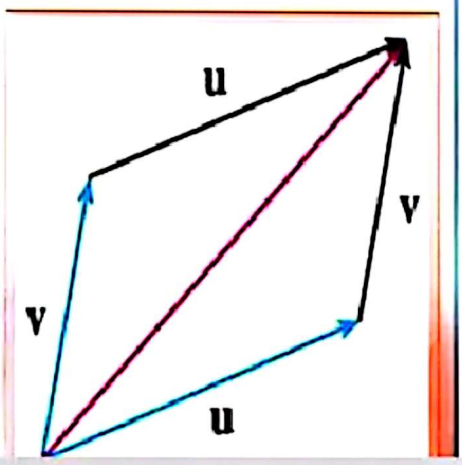
TRIANGLE LAW

You can see why this definition is sometimes called the Triangle Law.



VECTOR ADDITION

Here, we start with the same vectors **u** and **v** as earlier and draw another copy of **v** with the same initial point as **u**.



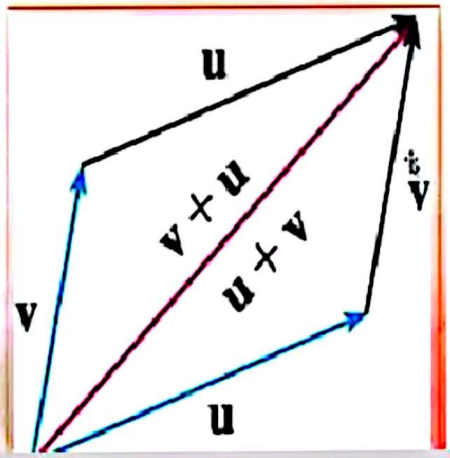
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VECTOR ADDITION

Completing the parallelogram,

we see that:

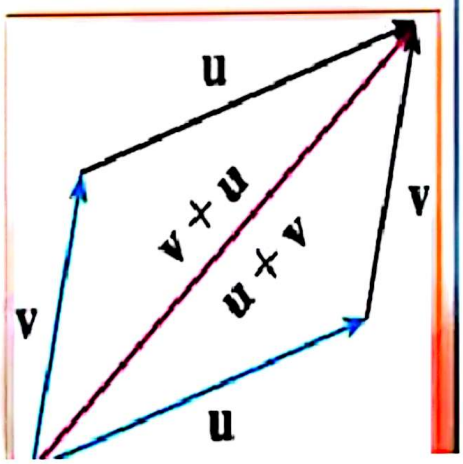
$$u + v = v + u$$



VECTOR ADDITION

This also gives another way to construct the sum:

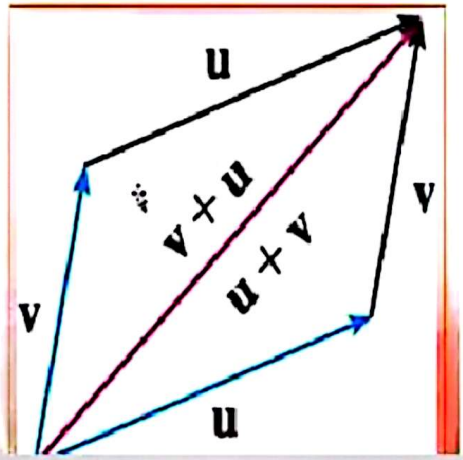
- If we place u and v so they start at the same point, then u + v lies along the diagonal of the parallelogram with u and v as sides.



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PARALLELOGRAM LAW

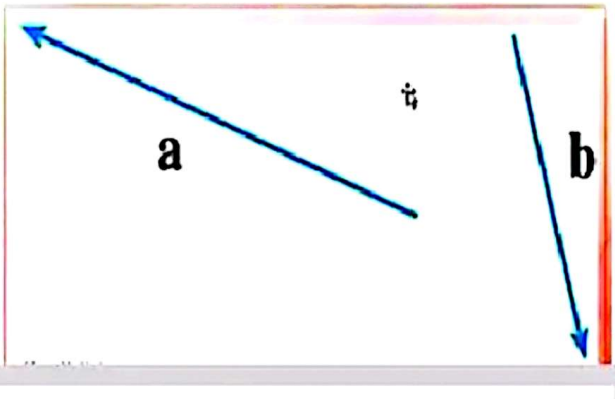
This is called the Parallelogram Law.



VECTOR ADDITION

Example 1

Draw the sum of the vectors a and b shown here.

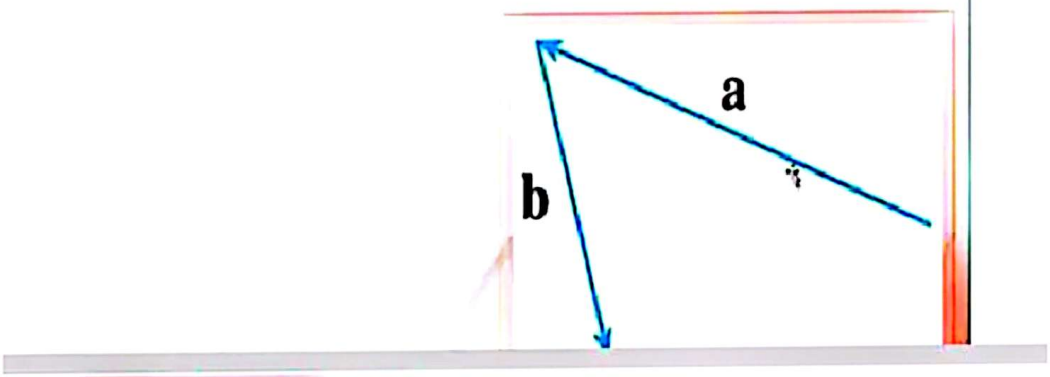


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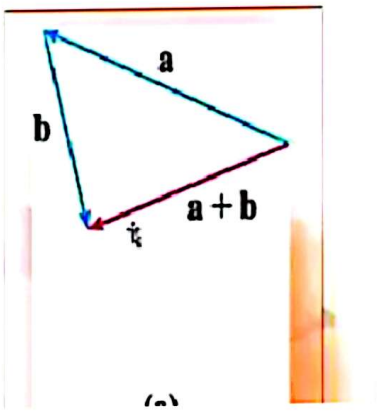
VECTOR ADDITION **Example 1**

First, we translate **b** and place its tail at the tip of **a**—being careful to draw a copy of **b** that has the same length and direction.



VECTOR ADDITION **Example 1**

Then, we draw the vector **a** + **b** starting at the initial point of **a** and ending at the terminal point of the copy of **b**.

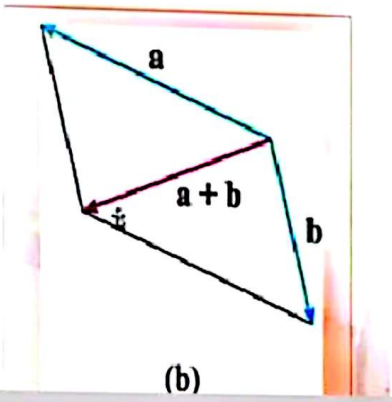


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VECTOR ADDITION

Example 1

Alternatively, we could place **b** so it starts where **a** starts and construct **a** + **b** by the Parallelogram Law.



MULTIPLYING VECTORS

It is possible to multiply a vector by a real number *c*.

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SCALAR

In this context, we call the real number c a scalar—to distinguish it from a vector.

MULTIPLYING SCALARS

For instance, we want $2\mathbf{v}$ to be the same vector as $\mathbf{v} + \mathbf{v}$, which has the same direction as $\mathbf{v}$ but is twice as long.

In general, we multiply a vector by a scalar as follows.

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If c is a scalar and $\mathbf{v}$ is a vector, the scalar multiple $c\mathbf{v}$ is:

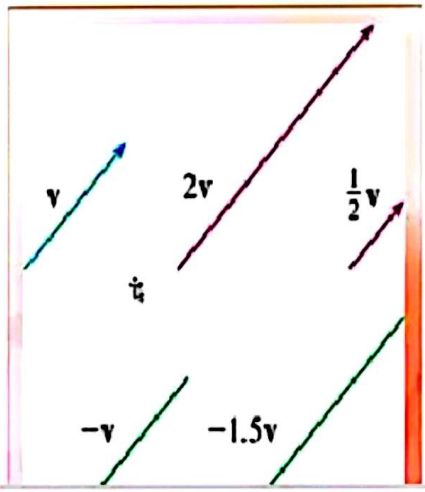
The vector whose length is $|c|$ times the length of $\mathbf{v}$ and whose direction is the same as $\mathbf{v}$ if $c > 0$ and is opposite to $\mathbf{v}$ if $c < 0$.

- If $c = 0$ or $\mathbf{v} = 0$, then $c\mathbf{v} = 0$.

SCALAR MULTIPLICATION

The definition is illustrated here.

- We see that real numbers work like scaling factors here.
- That's why we call them scalars.

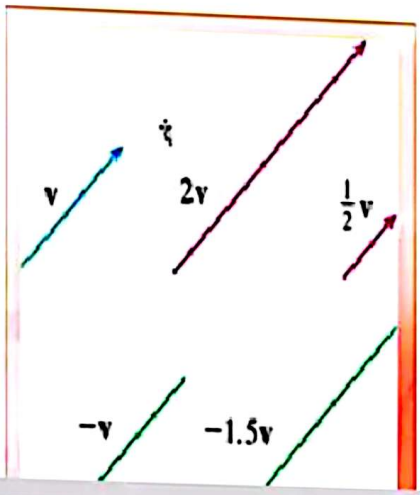


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SCALAR MULTIPLICATION

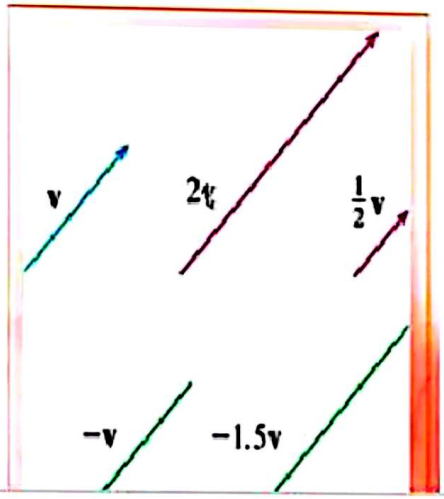
Notice that two nonzero vectors are parallel if they are scalar multiples of one another.



SCALAR MULTIPLICATION

In particular, the vector $-v = (-1)v$ has the same length as v but points in the opposite direction.

- We call it the negative of v.



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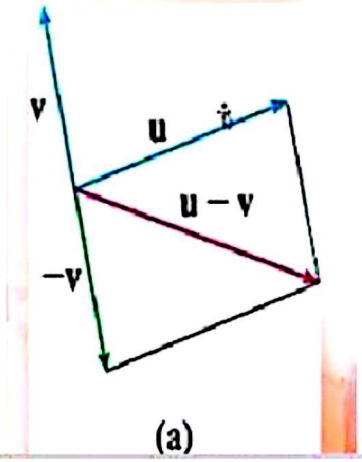
SUBTRACTING VECTORS

By the difference **u – v** of two vectors, we mean:

$$u - v = u + (-v)$$

SUBTRACTING VECTORS

So, we can construct **u – v** by first drawing the negative of **v**, **–v**, and then adding it to **u** by the Parallelogram Law.



(a)

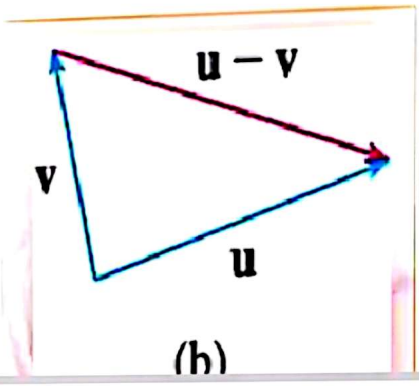
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SUBTRACTING VECTORS

Alternatively, since $\mathbf{v} + (\mathbf{u} - \mathbf{v}) = \mathbf{u}$, the vector $\mathbf{u} - \mathbf{v}$, when added to $\mathbf{v}$, gives $\mathbf{u}$.

SUBTRACTING VECTORS

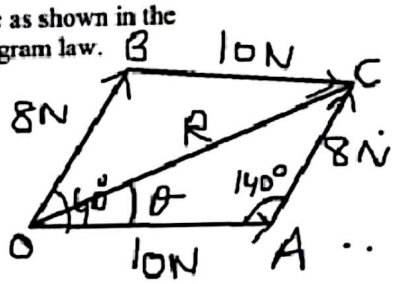
So, we could construct u by means of the Triangle Law.



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Example
Two forces of magnitude 8N and 10 N are acting on a particle as shown in the figure. Find the resultant of these forces by applying parallelogram law. Find also the angle made by the resultnat with 10N force.



Sol $\triangle OAC$

Law of Cosines

$$R^2 = 8^2 + 10^2 - 2 \times 8 \times 10 \times \cos 140^\circ$$
$$= 64 + 100 - 160 \cos 140^\circ$$
$$= 164 + 122.57$$
$$= 286.57$$

$$R = 17 \text{ N}$$

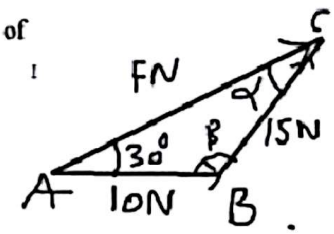
Law of Sines

$$\frac{R}{\sin 140^\circ} = \frac{8}{\sin\theta}$$
$$\frac{17}{\sin 140^\circ} = \frac{8}{\sin\theta}$$
$$\sin\theta = \frac{8}{17} \times \sin 140^\circ$$
$$= 0.3$$
$$\theta = \sin^{-1}(0.3)$$
$$\theta = 17.6^\circ$$

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Example

Two forces of magnitude 10N and 15N are the resolved parts of force of magnitude FN as shown in the figure. Find the magnitude of Force F.



Sol:

Using law of sines

$$\frac{15}{\sin 30^\circ} = \frac{10}{\sin\alpha}$$

$$\sin\alpha = \frac{10}{15}\times\sin 30^\circ = \frac{1}{3} \Rightarrow \alpha = \sin^{-1}\left(\frac{1}{3}\right) = 19.5^\circ$$

$$\beta = 180 - 30^\circ - 19.5^\circ$$

$$= 130.5^\circ$$

Now using law of cosines

$$F^2 = 10^2 + 15^2 - 2\times 10\times 15\cos 130.5^\circ$$

$= 130.5^\circ$

Now using law of cosines

$$F^2 = 10^2 + 15^2 - 2\times 10\times 15\cos 130.5^\circ$$

$$= 325 - 300\cos 130.5$$

$$= 325 + 194.83$$

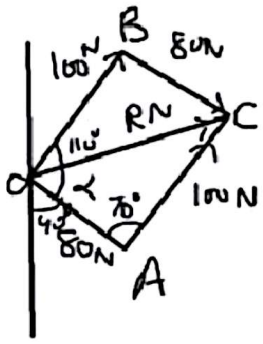
$$= 519.83$$

$$F = \sqrt{519.83}$$

$$\Rightarrow F = 22.8N$$

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Example
Two forces of magnitude 100N and 80N are acting on a particle as shown in the fig.
(i) Find the magnitude of the resultant by applying parallelogram law.
(ii) Find the angle that the resultant makes with vertical.



Sol

Using Law of cosines

$R^2 = 100^2 + 80^2 - 2\times100\times80\times\cos 70°$

$R = \sqrt{16{,}400 - 16000\cos 70°}$

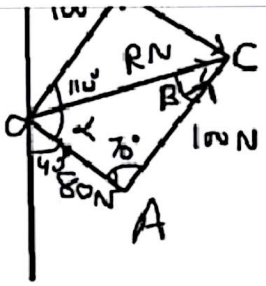
$= 104.54$

Using Law of Sines

$\frac{\sin\alpha}{80} = \frac{\sin 70°}{104.54}$

$\sin\alpha = \frac{80\sin 70°}{104.54}$

(ii) Find the angle that the resultant makes with vertical.



Sol

Using Law of cosines

$R^2 = 100^2 + 80^2 - 2\times100\times80\times\cos 70°$

$R = \sqrt{16{,}400 - 16000\cos 70°}$

$= 104.54\ N$

Using Law of Sines

$\frac{\sin\alpha}{80} = \frac{\sin 70°}{104.54}$

$\sin\alpha = \frac{80\sin 70°}{104.54}$

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$R = \sqrt{16{,}400 - 16000\cos 70^\circ}$

$= 104.54\ N$

Using Law of Sines

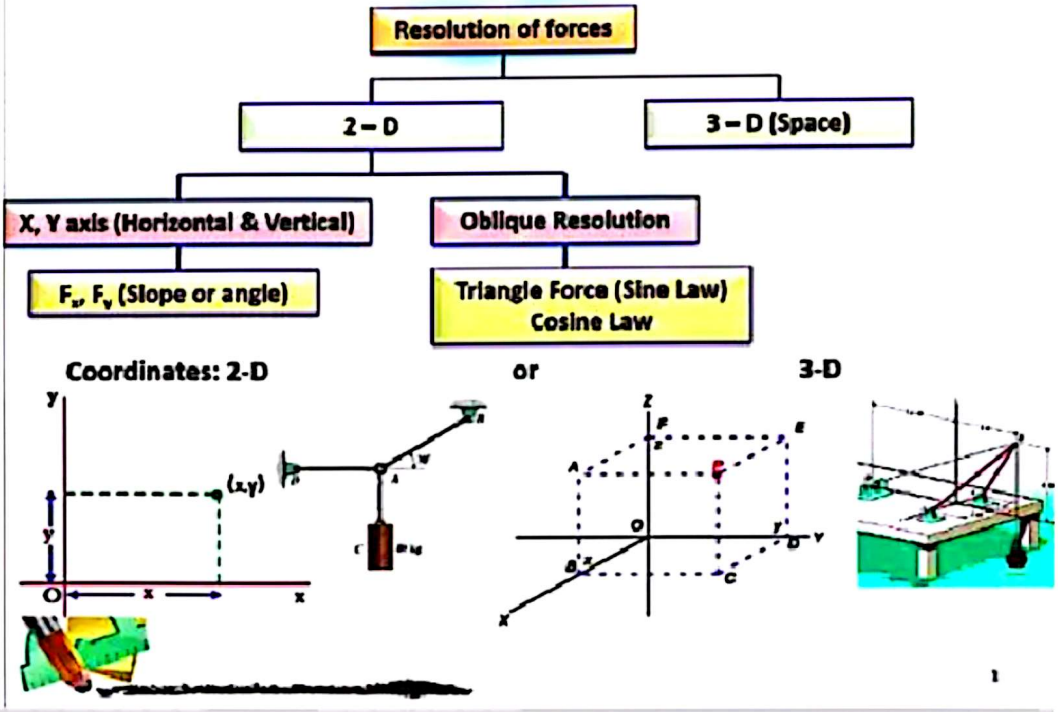
$\frac{\sin\alpha}{160} = \frac{\sin 70^\circ}{104.54}$

$\sin\alpha = \frac{160\sin 70^\circ}{104.54}$

$\alpha = 64^\circ$

Angle with vertical $= 64^\circ + 40^\circ = 104^\circ$

Resolution of Forces

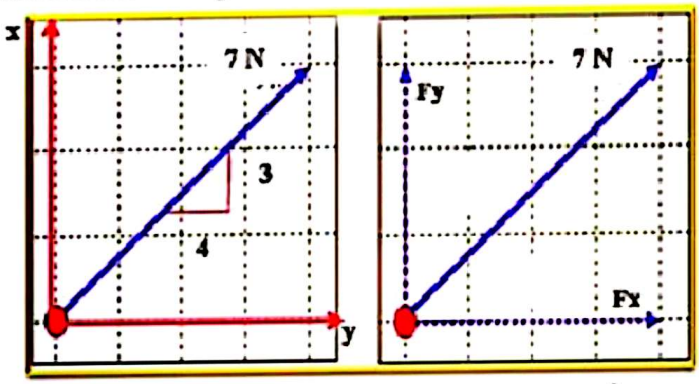


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Resolution of Forces

❖ The process of replacing a force system by its components is called ***Resolution***.

❖ To resolve a force into x-y components means to express the force as the sum of two forces, one in the x-direction (the x-component) and one in the y-direction (the y-component), When resolving a force into x-y components, we must have information on the direction of the force and the magnitude of the force.

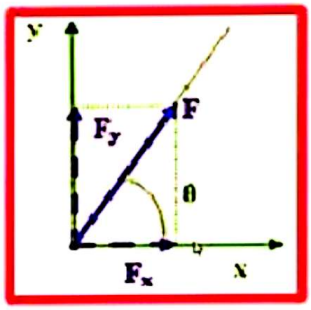


$F_x = 7 \times (\frac{4}{5})$
$F_y = 7 \times (\frac{3}{5})$

$F_x = 7 \times \cos 36.7^\circ$
$F_y = 7 \times \sin 36.7^\circ$

Resolution of Forces

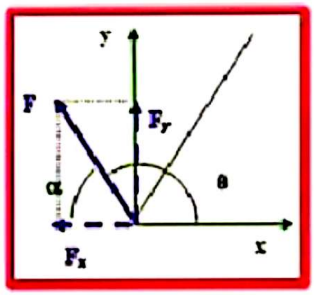
➢ When the **angle of the force relative to the x- or y-axes** is known, we can use trigonometry to find the components. Let (θ) be the angle that the force makes with the positive x-axis. Using trigonometry, we find the components (F_x and F_y) as follows:



$$\sin\theta = \frac{F_y}{F} \Rightarrow F_y = F.\sin\theta$$

$$\cos\theta = \frac{F_x}{F} \Rightarrow F_x = F.\cos\theta$$

➢ It is usually easiest to find the magnitudes of the components from the acute angle of the triangle defined by the force and the axes. Note that the components may have positive or negative signs. You need to recognize the signs of the components so they agree with their senses.



$$\sin\theta = \frac{F_y}{F} = \sin(180 - \alpha) \Rightarrow F_y = F.\sin\alpha$$

$$\cos\theta = \frac{F_x}{F} = \cos(180 - \alpha) \Rightarrow F_x = -F.\cos\alpha$$

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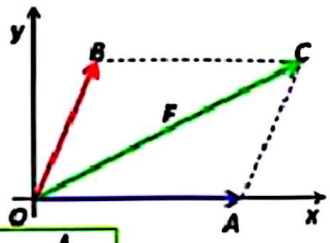
Resolution of Forces



Oblique Components (Oblique Forces):

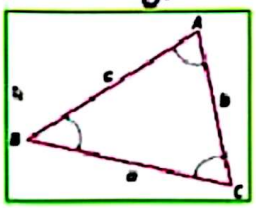
❖ If non-rectangular components of a force are needed, several methods are available for determining them.

❖ The components of the force (F) shown as (OA) and (OB) can be determined graphically by drawing the parallelogram to any convenient scale. The magnitudes of the components can be determined algebraically from the law of sines & cosines [With reference to triangle (ABC) with sides of (a, b, c), the sine rule states] which states:



Parallelogram Laws:

Sine Law: $$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

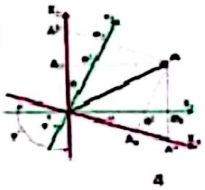


Cosine Law:

$$a^2 = b^2 + c^2 - 2\,b.c\,\cos A$$

$$b^2 = a^2 + c^2 - 2\,a.c\,\cos B$$

$$c^2 = a^2 + b^2 - 2\,a.b\,\cos C$$



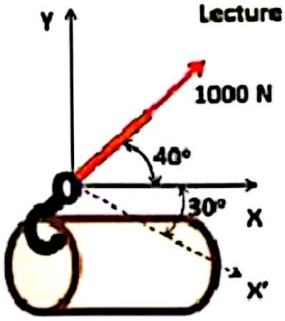


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Lecture 2

Example 1

❖ Resolve the (1000 N) force acting on the pipe, into components in the (a) X and Y directions, and (b) X' and Y directions.

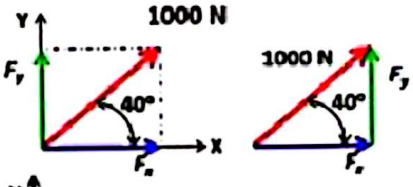


Solution:

Part (a): note that the length of the components is scaled along the X and Y axes by first constructing lines from the tip of (F = 1000 N) parallel to the axes in accordance with the parallelogram law.

$$F_x = 1000\cos 40° = 766\ N$$

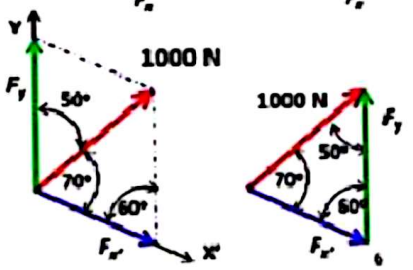
$$F_y = 1000\sin 40° = 643\ N$$



Part (b): note carefully how the parallelogram is constructed. Applying the law of sines using data mentioned in figure, it will yield:

$$\frac{F_{x'}}{\sin 50°} = \frac{1000}{\sin 60°} \rightarrow F_{x'} = 1000.\frac{\sin 50°}{\sin 60°} = 884.6\ N$$

$$\frac{F_y}{\sin 70°} = \frac{1000}{\sin 60°} \rightarrow F_y = 1000.\frac{\sin 70°}{\sin 60°} = 1085\ N$$



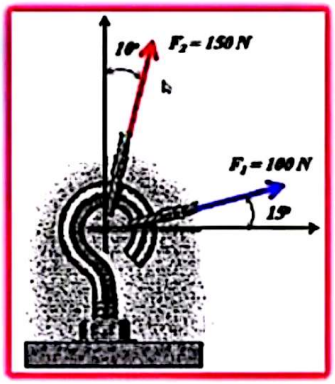


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Example 2

❖ The screw eye is subjected to two forces, F_1 and F_2. determine the magnitude and direction of the resultant force.



Solution:

Parallelogram Law: The parallelogram law of addition is shown in figure below. The two unknowns are the magnitude of (F_R) and the angle (θ).

Trigonometry: from the figure, the vector triangle is constructed. F_R is determined by using the law of cosines:

$$F_R = \sqrt{100^2 + 150^2 - 2(100)\cdot(150).\cos 115}$$

$$= \sqrt{10000 + 22500 - 30000\,(-0.4226)} = 212.6\,N$$

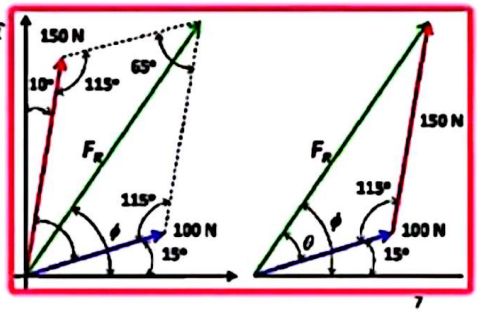
The angle (θ) is determined by applying the law of sines, using the computed value of (F_R):

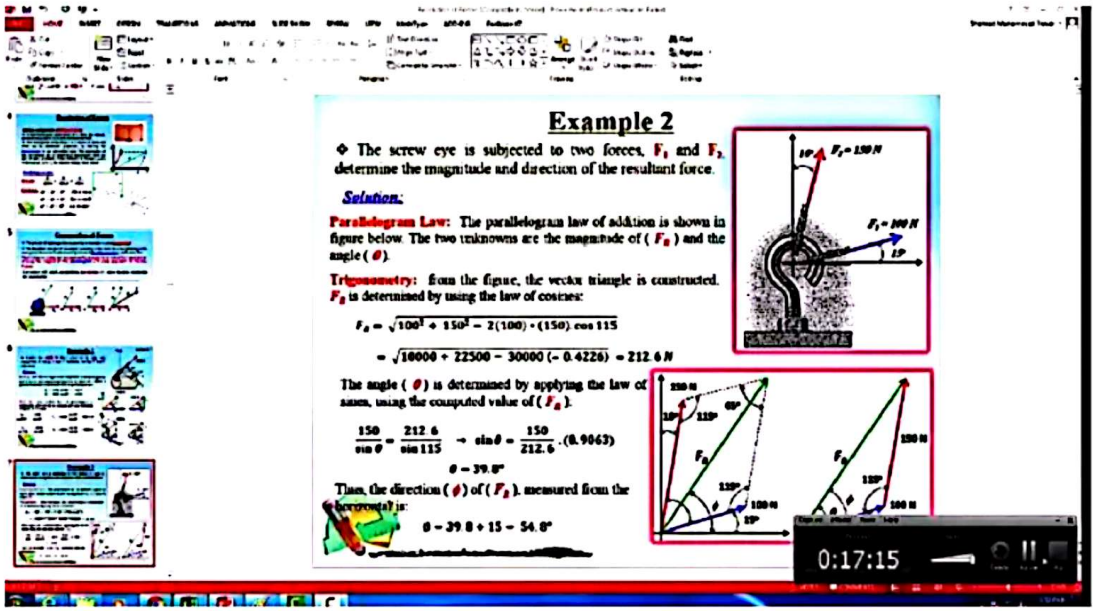
$$\frac{150}{\sin\theta} = \frac{212.6}{\sin 115} \rightarrow \sin\theta = \frac{150}{212.6}.(0.9063)$$

$$\theta = 39.8°$$

Thus, the direction (ϕ) of (F_R), measured from the horizontal is:

$$\emptyset = 39.8 + 15 = 54.8°$$

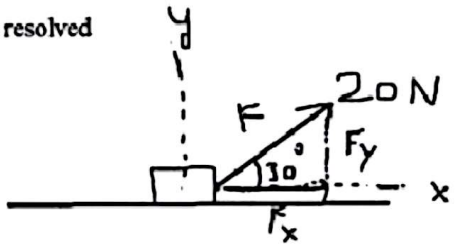




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Example
A force of 20N is acting on a block as shown in fig. Find its resolved parts along the coordinate axes.



Sol

$$F_x = F\cos\theta$$
$$= 20\cos 30^\circ$$
$$= 20\left(\frac{\sqrt{3}}{2}\right)$$
$$= 10\sqrt{3}\ N$$

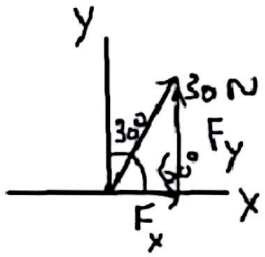
$$= 20\cos 30^\circ$$
$$= 20\left(\frac{\sqrt{3}}{2}\right)$$
$$= 10\sqrt{3}\ N$$

$$F_y = F\sin\theta$$
$$= 20\sin 30^\circ$$
$$= 20\left(\frac{1}{2}\right)$$
$$= 10N$$

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Example
a force of 30N is acting on the particle at an angle of 30 degrees with y-axis. Find the resolved parts along x-axis and y-axis.



Sol

Angle with x-axis $= 90° - 30°$
$= 60°$

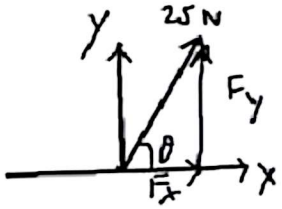
$F_x = F\cos\theta$
$= 30\cos 60°$
$= 30\left(\frac{1}{2}\right)$
$F_x = 15\ N$

$F_x = F\cos\theta$
$= 30\cos 60°$
$= 30\left(\frac{1}{2}\right)$
$F_x = 15\ N$

$F_y = F\sin\theta$
$= 30\sin 60°$
$= 30\left(\frac{\sqrt{3}}{2}\right)$
$F_y = 15\sqrt{3}\ N$

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Example
A force of 25N is acting on the particle as shown in the fig. Resolved part of force along x-axis is 15N. Find
(i) The angle that the force makes with x-axis.
(ii) Resolved part of the force along y-axis.



Sol:

$F_x = 15$ N

$F = 25$ N

(i) $\theta = ?$

(ii) $F_y = ?$

(ii) $F_y = ?$

$F_x = F \cos\theta$

$\cos\theta = \frac{F_x}{F}$

$\theta = \cos^{-1}\left(\frac{15}{25}\right)$

$= \cos^{-1}(0.6)$

$= 53$

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$$\theta = \cos^{-1}\left(\frac{15}{25}\right)$$

$$= \cos^{-1}(0.6)$$

$$= 53.13^\circ$$

$$F^2 = Fx^2 + Fy^2$$

$$25^2 = 15^2 + Fy^2$$

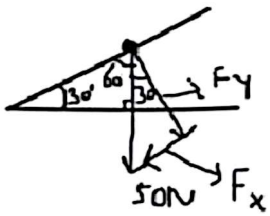
$$625 = 225 + Fy^2$$

$$Fy^2 = 400$$

$$Fy = 20\ N$$

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Example
A particle of mass 5kg is placed on an inclined plane as shown in the fig. Find the resolved parts of the weight of the particle in the direction parallel and perpendicular to the plane.



Sol:-

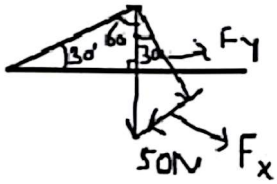
$m = 5\ kg$

$W = mg$

$= 5(10)$

$= 50N$

Sol:-



$m = 5\ kg$

$W = mg$

$= 5(10)$

$= 50N$

$F_x = mg \sin 30°$

$= 5(10) \sin 30°$

$= 50 \sin 30°$

$= 50\left(\frac{1}{2}\right)$

$= 25N$

E.

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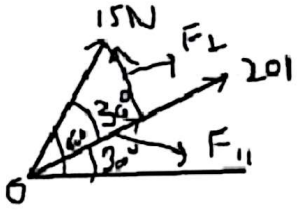
$$F_x = mg\sin 30^\circ$$
$$= 5(10)\sin 30^\circ$$
$$= 50\sin 30^\circ$$
$$= 50\left(\frac{1}{2}\right)$$
$$= 25\,N$$

$$F_y = F\cos 30^\circ = mg\cos 30^\circ$$
$$= 50\left(\frac{\sqrt{3}}{2}\right)$$
$$= 25\sqrt{3}\,N$$

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Example
A force of 15N and 20N are acting on the point as shown in the fig. Find the resolved component of the force of 15N in the direction of parallel and perpendicular to the force of 20N.



Sol:-

$$F_{||} = F\cos\theta$$
$$= 15\cos 30$$
$$= 15 \times \frac{\sqrt{3}}{2}$$
$$= 7.5\sqrt{3}\ N$$

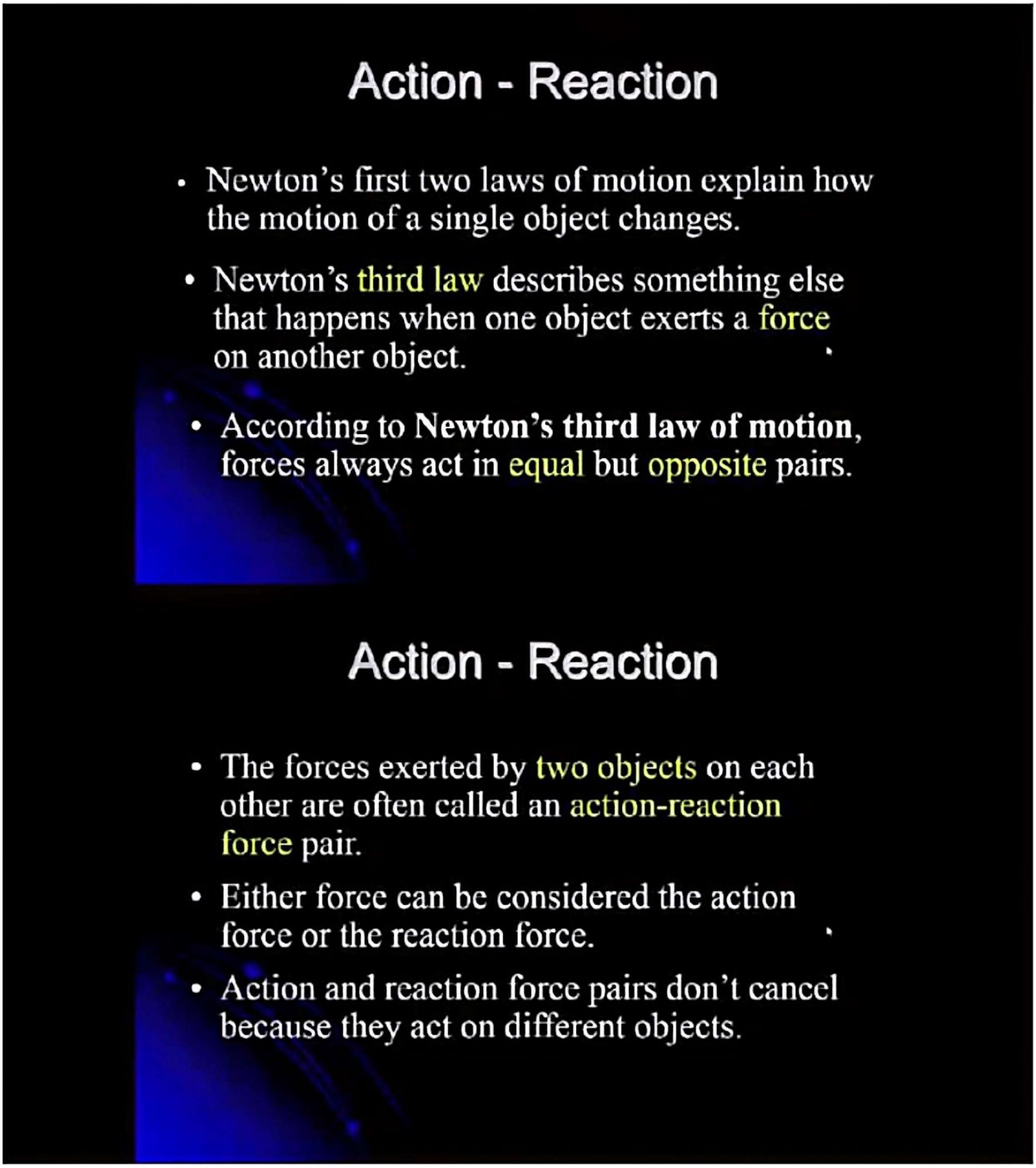


Sol:-

$$F_{||} = F\cos\theta$$
$$= 15\cos 30$$
$$= 15 \times \frac{\sqrt{3}}{2}$$
$$= 7.5\sqrt{3}\ N$$
$$F_{\perp} = F\sin\theta$$
$$= 15 \times \sin(30^\circ)$$
$$= 15\left(\frac{1}{2}\right)$$
$$= 7.5\ N$$

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Lecture No
07



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Newton's Third Law

- Newton's Third Law of Motion states: for every action, there is an equal but opposite reaction.
- This means that when you push on a wall, the wall pushes back on you with a force equal in strength to the force you exerted.

Action - Reaction

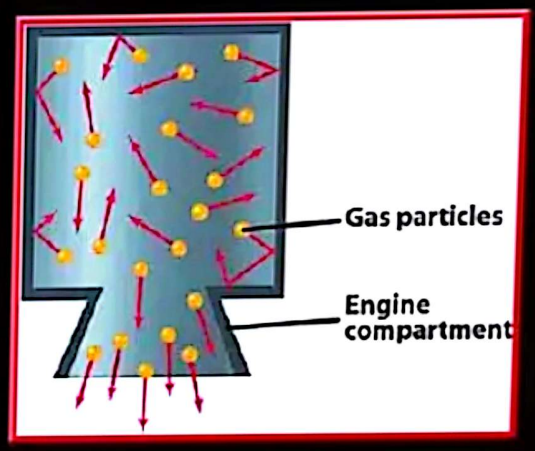
- You constantly use action-reaction force pairs as you move about.
- When you jump, you push down on the ground.
- The ground then pushes up on you. It is this upward force that pushes you into the air.



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Action - Reaction

- When the rocket fuel is ignited, a hot gas is produced.
- As the gas molecules collide with the inside engine walls, the walls exert a force that pushes them out of the bottom of the engine.



Action - Reaction

- This downward push is the action force.
- The reaction force is the upward push on the rocket engine by the gas molecules.
- This is the thrust that propels the rocket upward.

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